Project Report

Student Grade Management System

Course: Programming in Java

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1. Introduction

In school and college, teachers have to manage marks for many students. Calculating totals and giving grades manually can take a lot of time and lead to mistakes. This project is a simple Java-based tool that helps in recording marks and automatically calculating the final result for a student.

2. Problem Statement

Many teachers still use paper or simple spreadsheets to calculate grades. When there are many subjects, it becomes difficult to find the average and decide the correct grade quickly. This can be stressful during result season.

3. Objective

The main goal of this project is to create a Java application that:

- Takes student name and subject marks as input.
- Calculates the average percentage.
- Automatically assigns a grade (A, B, C, etc.).
- Displays a clear final report.

4. Methodology

The project is a console-based program written in Java.

- It uses a **Student Class** to store information.
- It uses an **Array** to store marks for different subjects.
- It uses a **Scanner** to take real-time input from the user.
- It uses **Logic (if-else)** to decide the grade based on the average score.

5. Features

- **Enter Subject Marks:** You can put marks for as many subjects as you want.
- **Auto-Grade:** The system decides the grade instantly.
- **Clean Summary:** Shows the name, average, and grade in a nice format.

6. Tools and Technologies Used

- **Language:** Java
- **Concepts:** Object-Oriented Programming (OOP), Arrays, Loops, and Conditional Statements.
- **IDE:** VS Code / IntelliJ IDEA.

7. Challenges Faced

- Setting up the logic to handle different numbers of subjects.
- Making sure the average calculation is accurate (using double for decimals).
- Designing a simple and easy-to-read output format.

8. Learning Outcomes

Through this project, I learned:

- How to use Classes and Objects to solve real-life problems.
- How to manage data using Arrays.
- How to take user input and show formatted output.

9. Future Improvements

- Add a way to save the data in a file so it doesn't delete when the program closes.

- Allow the system to manage an entire class of students at once.
- Add a Graphical User Interface (GUI) to make it look like a real app.

10. Conclusion

The **Student Grade Management System** is a small but useful tool. It shows how basic Java programming can make daily tasks easier and faster. It provides a good base for building bigger school management software in the future.

11. Demo

```
PS C:\Users\kirau> java GradeSystem
Enter marks for Subject 1: 69
Enter marks for Subject 2: 79
Enter marks for Subject 3: 89
Enter marks for Subject 4: 99
Enter marks for Subject 5: 43

--- Student Report ---
Name: Lokesh Kiraula
Average: 75.80
Final Grade: B
PS C:\Users\kirau>
```